Korra Tools Quick Start Guide

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Module: Module 08 - C Programming (Part II)
Layer: Deployment - Container
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📁 Required Files Checklist

Run every command from the `Fall 26 Files` folder unless a step says otherwise.

- ✅ `.env` (your API keys)
- ✅ `Dockerfile`
- ✅ `docker-compose.yml`
- ✅ `.dockerignore`
- ✅ `FA26_MD_06_S03_chatbot.py`
- ✅ `tools/file_stats_tool.py`
- ✅ `FA26_MD_07_file_stats/FA26_MD_07_file_stats/` (with its Makefile and tests)
- ✅ `FA26_MD_06_S04_container/FA26_MD_06_S04_container/requirements.txt`

🔑 .env File Format

```
OPENAI_API_KEY=sk-proj-xxxxxxxxxxxxxxxxxxxx
TAVILY_API_KEY=tvly-xxxxxxxxxxxxxxxxxxxx
```

No spaces around the `=`.

🧰 Before You Start

Do these three once, in order.

Confirm the Docker engine is running

```
docker ps
```

A table of columns means the engine is up. If this hangs or errors, Docker Desktop is not ready. `docker --version` is not enough; it answers even when the engine is down.

Install the Python packages

```
pip install -r "FA26_MD_06_S04_container\FA26_MD_06_S04_container\requirements.txt"
```

Build the C tool for Windows

```
cd tools
make
cd ..
```

This produces `file_stats.exe` in `tools`. Without it, step 1 below has nothing to run.

🧪 Test in Four Steps

Each step adds one layer. If a step fails, the layer it added is the problem.

1. The C tool by itself

From the `tools` folder:

```
.\file_stats.exe test.txt
```

Prints the JSON and exits with status `0`. This proves the compiled program works on its own, with no Python involved.

2. The C tool through the Python wrapper

From the `tools` folder:

```
python file_stats_tool.py test.txt
```

Same counts, now wrapped in `LangGraph Tool Result:`. This proves Python can find the executable and parse its JSON.

3. The C tool through the chatbot

From the `Fall 26 Files` folder:

```
python FA26_MD_06_S03_chatbot.py
```

Then ask:

```
You: Can you analyze the file tools/test.txt and tell me its statistics?
```

This proves the tool is bound to the agent and that the model chooses to call it.

4. The C tool through the chatbot in Docker

From the `Fall 26 Files` folder:

```
docker compose build
docker run --rm -it korra-phase-01
```

Then ask the same question:

```
You: Can you analyze the file tools/test.txt and tell me its statistics?
```

This proves the whole stack works in the container, where the C tool was compiled for Linux rather than Windows.

Every step reports the same counts for `test.txt`: **12** lines, **64** words, **379** characters, **379** bytes. If the numbers change between steps, the layer you just added is the one that broke them.

🚀 Everyday Commands

Once everything above passes, this is all you need day to day.

```
# Rebuild after changing the C tool or the chatbot
docker compose build

# Run Korra
docker run --rm -it korra-phase-01

# Run the C tool only, inside the container
docker run --rm korra-phase-01 python tools/file_stats_tool.py tools/test.txt
```

📞 Need Help?

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